PRODUCT REQUIREMENTS DOCUMENT

Trevolk AI Workforce

Section 1: Product Strategy & Foundation

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1. Product Overview

1.1 What is Trevolk AI Workforce?

Trevolk AI Workforce is a modern SaaS platform that lets businesses hire specialized AI Employees to handle repetitive business operations such as sales, customer support, appointment scheduling, and customer follow-ups. Each AI Employee is built to act as a digital team member — with defined responsibilities, workflows, memory, and business rules — rather than a generic chat window.

Multiple AI Employees operate together inside a single business workspace, sharing the same backend, authentication, database, and AI engine. This makes Trevolk a workforce of specialized AI roles, not a single all-purpose bot.

1.2 What Problem Does It Solve?

Businesses lose revenue and customer trust every day because of slow responses, missed follow-ups, and manual repetitive work. Trevolk solves this by giving businesses always-on AI Employees that qualify leads, answer customer questions, book appointments, and follow up automatically — without adding headcount.

1.3 Why Is It Different From Traditional Chatbots?

Traditional Chatbot	Trevolk AI Employee
Answers isolated questions	Owns an end-to-end business function (sales, support, scheduling, follow-up)
No persistent memory of the business or customer	Maintains memory, context, and business rules over time
Single generic use case	Multiple specialized employees working together in one workspace
Requires manual escalation for most tasks	Executes workflows (CRM updates, booking, reminders) and escalates only when needed
Add-on widget	Core operating layer of the business

1.4 Value Provided to Businesses

- Faster response times, which directly reduces lost leads
- Lower operational cost compared to hiring additional staff

- Consistent, always-available customer experience
- More organized sales and support workflows
- Time saved for founders and teams to focus on growth

2. Problem Statement

Small and mid-sized businesses run most of their sales and support operations manually. As customer volume grows, this manual approach breaks down, causing lost revenue and a poor customer experience.

2.1 Current Business Challenges

- Slow customer response times, especially outside business hours
- Leads going cold before a human can respond
- Missed or forgotten follow-ups on proposals and abandoned carts
- Manual, back-and-forth appointment booking
- Repetitive customer questions consuming staff time
- Inconsistent customer experience across channels
- Inefficient internal workflows with no central coordination

2.2 Manual Work Problems

Tasks like lead qualification, order tracking, and appointment scheduling are repetitive but still handled manually by staff. This limits how many customers a small team can serve and increases the risk of human error and delay.

2.3 Customer Communication Problems

Customers expect immediate answers across email, WhatsApp, and web chat. Businesses without dedicated round-the-clock staff cannot meet this expectation, which leads to frustrated customers and lost sales opportunities.

2.4 Sales and Operational Inefficiencies

Without a system to qualify leads and manage follow-ups consistently, sales teams waste time on unqualified prospects while forgetting to circle back with high-intent leads. This directly reduces conversion rates and revenue.

3. Target Market

3.1 Primary Target Industries

- E-commerce Businesses
- Digital Agencies
- Software Houses
- Online Coaches & Educators

- Real Estate Agencies

3.2 Ideal First Customers

The ideal first customers are small to mid-sized digital-first businesses that already handle a meaningful volume of leads or customer conversations, but do not yet have the budget or need for a full in-house team. These businesses feel the pain of missed leads and slow response times most acutely, and can adopt a new tool quickly since they typically have lean, tech-comfortable teams.

3.3 Why These Industries Need AI Employees

Industry	Why They Need AI Employees
E-commerce	High volume of repetitive order, shipping, and return questions; abandoned carts need fast recovery
Digital Agencies	Constant inbound leads need fast qualification and scheduling to avoid losing clients to competitors
Software Houses	Pre-sales questions and meeting scheduling take up valuable technical team time
Online Coaches & Educators	Need consistent follow-up and appointment booking without hiring a support team
Real Estate Agencies	Property inquiries require fast response and appointment scheduling to convert leads

3.4 First Market to Target, and Why

E-commerce businesses and digital agencies should be the first market focus. Both segments have high, measurable pain around lead response time and repetitive customer questions, are comfortable adopting SaaS tools, and provide a clear, demonstrable ROI story (recovered carts, faster lead response) that supports early sales and case studies.

4. Customer Personas

4.1 Persona 1: Ayesha — E-commerce Store Owner

- Business type: Small online fashion store, 2-person team
- Problems: Cannot answer order and shipping questions fast enough; loses sales from abandoned carts
- Goals: Respond to customers instantly and recover more abandoned carts
- How Trevolk helps: AI Customer Support Employee handles order tracking and FAQs; AI Follow-up Employee recovers abandoned carts automatically

4.2 Persona 2: Bilal — Digital Agency Founder

- Business type: Small digital marketing agency, handles inbound leads from ads and website

- Problems: Leads go cold before his sales team can respond; manual scheduling wastes time
- Goals: Qualify and respond to leads faster, book more discovery calls
- How Trevolk helps: AI Sales Employee qualifies leads and updates CRM; AI Receptionist books meetings automatically

4.3 Persona 3: Sarah — Online Coach

- Business type: Solo online business coach selling courses and coaching calls
- Problems: No time to answer repetitive questions or chase unresponsive leads
- Goals: Stay focused on coaching while not losing potential clients
- How Trevolk helps: AI Follow-up Employee re-engages leads; AI Receptionist manages her calendar and bookings

4.4 Persona 4: Omar — Real Estate Agent

- Business type: Independent real estate agency with a small team
- Problems: Property inquiries need fast answers; missed calls mean missed viewings
- Goals: Respond to every inquiry quickly and keep his calendar full of viewings
- How Trevolk helps: AI Receptionist books and reschedules viewings; AI Sales Employee answers pre-sales questions and notifies Omar of hot leads

4.5 Persona 5: Hassan — Software House Co-founder

- Business type: Small software development company handling client inquiries
- Problems: Developers get pulled into answering repetitive pre-sales questions and scheduling calls
- Goals: Keep the technical team focused on delivery, not admin work
- How Trevolk helps: AI Sales Employee handles pre-sales questions and meeting scheduling; AI Customer Support Employee manages client inquiries

5. Product Vision & Goals

5.1 Short-Term Goal

Launch an MVP with a focused set of AI Employees (Sales, Customer Support, Receptionist, Follow-up) that can be deployed by small businesses in e-commerce and digital agencies within their first week of signup.

5.2 Long-Term Vision

Build the most intelligent AI Workforce platform that allows businesses of all sizes to automate repetitive operations through specialized AI Employees, eventually expanding into a full marketplace of AI Employees across departments such as HR, marketing, finance, and operations.

5.3 MVP Objective

Prove that a business can replace manual, repetitive sales and support work with AI Employees that operate reliably inside a single workspace, and demonstrate measurable time and cost savings for early customers.

5.4 Success Criteria

- Early customers report faster response times and fewer missed leads
- AI Employees handle a majority of repetitive inquiries without human escalation
- Businesses activate more than one AI Employee after onboarding
- Positive retention and renewal from early paying customers
- Clear, repeatable onboarding process from signup to active AI Employee

6. Product Scope

6.1 MVP Scope

AI Employee	MVP Responsibilities
AI Sales Employee	Qualify leads, answer pre-sales questions, schedule meetings, update CRM, notify sales team
AI Customer Support Employee	Answer FAQs, handle inquiries, track orders, manage complaints, process returns, escalate when needed
AI Receptionist	Book appointments, manage calendars, reschedule meetings, send reminders, handle availability
AI Follow-up Employee	Lead follow-ups, proposal reminders, abandoned cart recovery, re-engagement, email & WhatsApp campaigns

6.2 Future Scope

- AI HR Employee
- AI Marketing Employee
- AI Finance Employee
- AI Operations Employee
- AI Recruitment Employee
- Expansion into clinics, restaurants, SMEs, and enterprise accounts
- Marketplace for third-party or add-on AI Employees

The MVP intentionally stays limited to four AI Employees and core business-critical workflows. Additional employees and industries are deferred until the platform proves value with its first customer segments.

7. Unique Value Proposition

7.1 Why Businesses Choose Trevolk

- A workforce of specialized AI Employees instead of one generic assistant
- Employees that own full workflows end-to-end, not just answer questions
- Unified workspace so all AI Employees share context and infrastructure
- Enterprise-grade experience built for growing businesses, not just enterprises

7.2 AI Workforce vs. Normal Automation Tools

Normal Automation Tools	Trevolk AI Workforce
Rule-based, rigid workflows	AI-driven decisions that adapt to context and conversation
Single-purpose tools stitched together	Multiple specialized employees working from one shared workspace
Requires manual setup for every workflow	Employees come with pre-built responsibilities for their role
No shared memory across tools	Shared memory and business rules across all AI Employees

7.3 Competitive Advantage

Trevolk's advantage is being built as a unified AI Workforce from the start — one platform, one workspace, and one AI engine powering multiple specialized employees — rather than a collection of disconnected point solutions. This makes it easier for businesses to adopt, and harder for competitors to replicate quickly.

8. Business Model Overview

8.1 SaaS Model

Trevolk operates as a cloud-based, multi-tenant SaaS platform. Businesses sign up, activate the AI Employees relevant to their needs, and pay a recurring subscription based on usage and the number of active employees.

8.2 Subscription Approach

- Monthly subscription for flexibility
- Annual subscription at a discount to improve retention and cash flow
- Pricing scaled by number of active AI Employees and usage volume

8.3 Possible Pricing Tiers

Tier	Target Customer	Includes
Starter	Solo founders, small teams	1–2 AI Employees, core workflows, standard support
Growth	Growing SMEs and agencies	All 4 MVP AI Employees, integrations, priority support
Enterprise	Larger businesses, multi-location teams	Custom AI Employees, dedicated support, advanced integrations

8.4 Enterprise Opportunities

As the platform matures, Enterprise plans will offer custom AI Employees, dedicated onboarding, and advanced integrations for larger organizations. Longer term, a marketplace for AI Employees can open additional revenue streams beyond core subscriptions.

End of Section 1: Product Strategy & Foundation — to be merged with subsequent PRD sections.

SECTION 2

User Experience (UX) & UI Design Requirements

Product Requirements Document — Trevolk AI Workforce

1. Product Experience Overview

1.1 UX Philosophy

Trevolk AI Workforce is experienced as a digital company, not a chat widget. Every design decision should reinforce one idea: the business owner is managing a team of AI Employees, not configuring a bot.

- Clarity over cleverness — every screen has one primary purpose.
- Status-driven design — the business owner should always know what each AI Employee is doing, right now.
- Low cognitive load — enterprise-grade power presented through a simple, minimal interface.
- Trust through transparency — every AI action is visible, logged, and explainable.
- Fast to value — a new business should be able to activate one AI Employee and go live within a single onboarding session.

1.2 How Businesses Interact With The Platform

Businesses interact with Trevolk on two layers:

- Management Layer (Dashboard): the business owner and team configure, monitor, and control AI Employees.
- Delivery Layer (Business Channels): AI Employees interact with customers through the business's own channels — website chat, WhatsApp, email, calls, or forms — not through a separate Trevolk-branded app.

The dashboard is where the business works. The AI Employees are where the customer experiences the business.

1.3 Chatbot Experience vs. AI Employee Platform Experience

Dimension	Traditional Chatbot	Trevolk AI Employee Platform
Mental model	A script answering FAQs	A team member with a role, memory, and responsibilities
Scope	Single, isolated conversation flow	Multiple specialized employees working together
Visibility	Conversation log only	Full activity history, performance metrics, and outcomes
Control	Edit a flow / script	Configure behavior, rules, working hours, escalation logic
Trust	Black box responses	Transparent status, actions taken, and reasoning surfaced to the owner
Growth	Add more flows	Activate more AI Employees from a shared workforce

2. User Types & Their Experience

2.1 Business Owner

The primary decision-maker and account admin. Needs a command-center view of the entire AI Workforce.

- Needs to see: overall performance, revenue/lead impact, AI Employee status (active/inactive), alerts requiring attention, billing/plan status.
- Needs to manage: activating/deactivating AI Employees, team member access, integrations, business rules, subscription plan.
- Primary emotion to design for: confidence and control — the owner should feel the AI Workforce is working for them, not operating unpredictably.

2.2 Team Member

Sales reps, support agents, and operations staff who collaborate with AI Employees day-to-day rather than configure the whole workspace.

- Role-based dashboard access — sees only the AI Employees and modules relevant to their function (e.g., a sales rep sees Leads and Conversations, not billing).
- Collaboration surface: hand-off queue for conversations escalated by an AI Employee, ability to reply inline, ability to leave internal notes.
- Notifications for tasks needing human attention (new lead assigned, escalation, appointment conflict).

2.3 Customer

The end customer never logs into Trevolk. They experience the AI Employee entirely inside the business's own channels.

- Interacts via the channel the business already uses: website chat widget, WhatsApp, email, or phone/voice (future).
- Experience should feel like speaking with a helpful, responsive staff member — natural language, fast replies, no visible 'AI platform' branding unless the business chooses to disclose it.
- Clear path to human hand-off at any point if the AI Employee cannot resolve the request.

3. Complete User Journey

3.1 Business Side Journey

Landing Page → Signup → Create Workspace → Select AI Employees → Setup → Connect Integrations → Go Live → Monitor Performance

Stage	What Happens	Design Priority
Landing Page	Visitor learns what an AI Employee is and picks a use case	Clarity of value proposition, low-friction CTA
Signup	Account creation (email or SSO via Clerk/Supabase)	Minimal fields, fast completion

Stage	What Happens	Design Priority
Create Workspace	Business profile set up (name, industry, branding)	Guided, single-flow wizard
Select AI Employees	Owner chooses which AI Employees to activate	Clear role cards, easy comparison
Setup	Configure business rules, knowledge base, working hours	Step-by-step, progress indicator
Connect Integrations	Link WhatsApp, Calendar, CRM, Stripe, etc.	One-click connect where possible, clear status
Go Live	AI Employee activated on live channels	Confidence-building confirmation state
Monitor Performance	Ongoing use of dashboard, analytics, adjustments	Always-visible status and outcomes

3.2 Customer Side Journey

Customer Query → AI Employee Interaction → AI Action → Business Update → Resolution

Stage	What Happens
Customer Query	Customer reaches out via chat, WhatsApp, or email with a question or request
AI Employee Interaction	The relevant AI Employee responds using business knowledge, memory, and rules
AI Action	AI Employee performs an action: books an appointment, updates CRM, tracks an order, escalates
Business Update	Dashboard reflects the new lead, conversation, or appointment in real time
Resolution	Query resolved by AI, or handed off to a human team member with full context

4. Website Structure (Public Pages)

Public-facing marketing site required to acquire and convert business customers.

Page	Goal	Important Content	Main CTA
Homepage	Communicate the AI Workforce concept in seconds	Hero value prop, AI Employee overview, social proof, how-it-works	Start Free Trial
Solutions	Show how Trevolk solves specific business problems	Problem/solution pairs (lost leads, missed follow-ups, etc.)	See How It Works
AI Employees	Introduce each AI Employee as a distinct product	Role cards for Sales, Support, Receptionist, Follow-up	Explore AI Employees
Industries	Show relevance to target verticals	E-commerce, Agencies, Software Houses, Coaches, Real Estate	View Industry Fit

Page	Goal	Important Content	Main CTA
Pricing	Enable self-serve plan comparison	Monthly/Annual toggle, plan feature comparison, Enterprise contact	Choose Plan
About	Build trust and credibility	Vision, mission, team, story	Contact Us
Contact	Convert high-intent or enterprise visitors	Contact form, demo request, support email	Book a Demo
Login	Return access for existing users	Email/SSO login, password reset	Log In
Signup	Convert new visitors into workspaces	Simple signup form, SSO options	Create Workspace

5. Dashboard Structure

5.1 Sidebar Navigation

- Dashboard
- AI Employees
- Conversations
- Customers
- Leads
- Appointments
- Knowledge Base
- Automations
- Analytics
- Integrations
- Settings

Sidebar is collapsible, icon-first, with active-state highlighting in Primary Accent Blue.

5.2 Main Dashboard Sections & Widgets

Section	Purpose	Key Widgets / Cards
Dashboard (Home)	At-a-glance health of the whole AI Workforce	AI Employee status cards, today's activity summary, alerts, quick actions
AI Employees	Manage each employee individually	Employee cards with status toggle, performance snapshot, configure button
Conversations	Unified inbox across all channels	Channel filter, conversation list, escalation flags, AI/human indicator
Customers	Central customer record	Customer profile, interaction history, linked orders/appointments

Section	Purpose	Key Widgets / Cards
Leads	Sales pipeline visibility	Lead list/kanban, qualification status, source, assigned AI Employee
Appointments	Calendar and booking overview	Calendar view, upcoming appointments, reschedule/cancel actions
Knowledge Base	Source of truth AI Employees draw from	Document/FAQ list, upload area, sync status
Automations	Rules and workflow configuration	Automation list, trigger/action builder, active/inactive toggle
Analytics	Business outcomes and AI performance	Response time, resolution rate, conversion rate, revenue impact charts
Integrations	Connected tools	Integration cards (WhatsApp, Calendar, Stripe, Slack, etc.), connect/disconnect
Settings	Workspace and account configuration	Business profile, team & roles, billing, notifications

5.3 Common User Actions

- Toggle an AI Employee active/inactive
- Jump into a live or past conversation
- Reassign or escalate a conversation to a team member
- Approve or edit an AI-suggested action (e.g., a reply or booking)
- Add/edit knowledge base content
- Invite a team member and assign a role
- Review analytics and export reports

6. AI Employee Screens

Each AI Employee has a consistent screen template so the experience feels unified across the workforce, while its content reflects its specific responsibilities.

6.1 Shared Screen Template

Screen Element	Description
Overview Screen	Role summary, current status, quick stats, and a start/pause control
Status	Active / Paused / Needs Setup / Needs Attention — shown as a colored badge
Configuration Options	Business rules, tone of voice, working hours, escalation thresholds, connected knowledge base
Performance Metrics	Volume handled, resolution/conversion rate, response time, outcome trend

Screen Element	Description
Activity History	Chronological log of conversations and actions taken by this AI Employee
Controls	Pause/resume, edit configuration, view logs, escalate settings, retrain/update knowledge

6.2 AI Sales Employee

- Overview: leads qualified this period, meetings scheduled, CRM sync status.
- Configuration: qualification criteria, meeting scheduling rules, CRM field mapping.
- Metrics: lead-to-meeting conversion rate, average qualification time, notifications sent.
- Controls: adjust qualification questions, pause outreach, manual lead review queue.

6.3 AI Customer Support Employee

- Overview: open tickets/conversations, FAQ coverage, escalation rate.
- Configuration: FAQ/knowledge source, order tracking connection, return/complaint workflow, escalation rules.
- Metrics: resolution rate, average response time, customer satisfaction (if collected).
- Controls: edit FAQ responses, adjust escalation triggers, review flagged complaints.

6.4 AI Receptionist

- Overview: today's appointments, calendar sync status, upcoming reschedules.
- Configuration: calendar connection, availability rules, reminder timing.
- Metrics: bookings made, reschedule rate, no-show rate, reminder delivery rate.
- Controls: block/unblock availability, edit reminder templates, manual booking override.

6.5 AI Follow-up Employee

- Overview: active follow-up sequences, re-engagement rate, recovered carts/proposals.
- Configuration: follow-up sequence timing, channel selection (email/WhatsApp), trigger events.
- Metrics: response/reply rate, recovery rate, revenue recovered.
- Controls: pause a sequence, edit message templates, view per-customer follow-up timeline.

7. Design System Guidelines

7.1 Visual Style

- Modern enterprise SaaS aesthetic — comparable in polish to Linear, Vercel, or Notion.
- Minimal, spacious layouts with strong content hierarchy.
- Soft rounded corners and subtle depth (light shadows, minimal glass effects) instead of heavy skeuomorphism.
- Professional and calm rather than colorful or playful — the interface should feel like enterprise infrastructure, not a consumer app.

7.2 Color Palette

Palette follows the Master Context Document exactly; no new colors should be introduced.

Role	Color	Usage
Primary Background	Dark Charcoal	App shell, sidebar, headers
Primary Accent	Blue	Primary buttons, active states, links
Secondary Accent	Sky Blue	Secondary highlights, chart accents
Text	White / Light Gray	Body and heading text on dark surfaces
Success	Green	Positive status, completed actions
Warning	Orange	Needs-attention states
Danger	Red	Errors, failed actions, critical alerts

7.3 Typography

- A clean, modern sans-serif typeface (e.g., Inter, Satoshi, or system-equivalent) for both UI and marketing content.
- Clear type scale: distinct sizes for page titles, section headings, card titles, body text, and captions.
- Generous line height for readability in dense dashboard views.
- Numeric/metric data may use tabular figures for alignment in tables and charts.

7.4 Core Components

Component	Design Notes
Cards	Rounded corners, subtle border or shadow, used for AI Employee status, metrics, and list items
Buttons	Primary (filled blue), Secondary (outline), Ghost/text — consistent height and radius across the app
Tables	Zebra-striped rows, sticky header, inline row actions, sortable columns where relevant
Forms	Single-column layout, inline validation, clear required-field indicators
Charts	Line/bar charts for trends, donut/progress for rates — accent colors reserved for key data series
Modals	Centered, focused single-task dialogs with clear primary/secondary actions
Navigation	Persistent collapsible sidebar + top bar for workspace switcher, search, and notifications

8. Animation & Interaction Guidelines

Animation exists to support usability and communicate system status — never for decoration alone.

Interaction	Guideline
Page Transitions	Smooth, brief fade/slide between views (150–250ms); no jarring cuts
Loading States	Skeleton screens for dashboard data; spinners reserved for short, blocking actions
Skeleton Screens	Match the shape of the content being loaded (cards, table rows, charts)
Hover Effects	Subtle elevation or color shift on cards and buttons to indicate interactivity
Micro-interactions	Button press feedback, toggle animations, success checkmarks on save
Sidebar Behavior	Smooth expand/collapse transition; active item highlighted with accent color
Typing Indicators	Used in live conversation views to show an AI Employee is composing a response
Empty States	Friendly illustration or icon + short explanation + clear next action (e.g., 'Connect your first integration')

Rule of thumb: if an animation cannot be justified by clarity, feedback, or perceived performance, it should not be included.

9. Responsive Design

Breakpoint	Sidebar Behavior	Dashboard Adaptation	Navigation
Desktop (≥1200px)	Persistent, expanded sidebar	Full multi-column layout, all widgets visible	Sidebar + top bar
Tablet (768–1199px)	Collapsible to icon-only sidebar	Widgets reflow to fewer columns, tables scroll horizontally	Collapsible sidebar + top bar
Mobile (<768px)	Hidden by default, opens as overlay drawer	Single-column stacked cards, condensed tables become lists	Bottom nav or hamburger menu

The public marketing site follows standard responsive best practices (fluid grid, mobile-first navigation) and is out of scope for the dashboard breakpoints above.

10. UX Principles

Principle	Rule
Simplicity	Every screen should have one clear primary action; secondary actions stay visually secondary
Accessibility	Sufficient color contrast on dark backgrounds, keyboard navigability, readable font sizes, alt text for icons/illustrations
Fast Interactions	Perceived performance prioritized via skeleton states and optimistic UI updates where safe
Clear Feedback	Every user action (save, connect, activate) produces an immediate, visible confirmation
Error Handling	Errors are explained in plain language with a clear next step, never a raw technical message

Principle	Rule
User Confidence	AI actions are always visible and explainable, so the business owner always understands what happened and why

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End of Section 2 — User Experience (UX) & UI Design Requirements. To be merged with other PRD sections covering backend architecture, database design, APIs, AI agent implementation, and deployment.

Section 3: AI Employees & Business Workflows

This section defines the product behavior of every AI Employee in the Trevolk AI Workforce MVP: what each employee does, how it behaves, the workflow it follows, the rules that bound its autonomy, and how its performance is measured. It is written at the product and workflow level. UI, backend architecture, database schema, and deployment are covered in other sections of this PRD.

3.1 AI Workforce Overview

What Is an AI Employee?

An AI Employee is a purpose-built AI system assigned to a specific business function — sales, support, reception, or follow-up — with its own responsibilities, workflows, business rules, and memory. It is designed and evaluated the way a real team member would be: by the work it completes and the outcomes it drives, not by the conversations it holds.

How Is It Different From a Chatbot?

Dimension	Traditional Chatbot	Trevolk AI Employee
Scope	Answers isolated questions	Owns an end-to-end business function
Memory	Little to no persistent memory	Remembers customers, history, and context
Action	Only replies with text	Takes real actions: books, updates, notifies
Rules	Generic, scripted responses	Follows business-specific rules and limits
Accountability	No ownership of outcomes	Measured on business outcomes (KPIs)

How Multiple AI Employees Work Together

All AI Employees operate inside a single business workspace and share the same customer records, conversation history, and business context. This shared context allows employees to hand off work to one another without the customer having to repeat themselves. For example, the AI Sales Employee can qualify a lead and pass the confirmed meeting time to the AI Receptionist for booking, or the AI Customer Support Employee can flag a dissatisfied customer for the AI Follow-up Employee to re-engage later. Each employee owns a distinct responsibility, but together they behave as one coordinated workforce rather than separate disconnected tools.

Value AI Employees Provide to Businesses

- Faster response times across sales, support, and scheduling
- Fewer lost leads and missed follow-ups
- Lower operational cost compared to hiring for repetitive roles
- Consistent customer experience, 24/7, without added headcount
- Business owners regain time spent on repetitive coordination work

3.2 AI Sales Employee

Purpose

Businesses lose revenue when leads are not responded to quickly, are never qualified, or fall through the cracks before a human ever engages them. The AI Sales Employee exists to respond to every incoming lead immediately, determine whether the lead is worth the sales team's time, and move qualified leads toward a booked meeting.

Responsibilities

- Handle incoming leads from forms, chat widgets, or connected channels
- Ask qualification questions (budget, need, timeline, authority)
- Understand customer intent and interest level
- Recommend next actions to the customer (e.g., book a call, view pricing)
- Schedule meetings directly with the sales team's availability
- Update CRM records with lead details and qualification status
- Notify the sales team when a lead is qualified or requests a human

Workflow

Step	Stage	What Happens
1	**New Lead**	A lead arrives through a connected channel (website form, WhatsApp, chat widget) and is assigned to the AI Sales Employee.
2	**Conversation**	The AI Sales Employee greets the lead and begins a natural conversation to understand what they need.
3	**Qualification**	The employee asks qualifying questions to assess fit, budget range, timeline, and decision-making authority.
4	**Lead Score**	Based on the conversation, the lead is scored (e.g., Hot / Warm / Cold) using business-defined qualification criteria.
5	**CRM Update**	Lead details, responses, and score are written to the CRM automatically.
6	**Meeting Booking**	For qualified leads, the employee offers available time slots and books a meeting.

Step	Stage	What Happens
7	**Follow-up**	The lead is handed to the AI Follow-up Employee if no meeting is booked, or to the sales team once booked.

Business Rules

Can Do	Cannot Do	Escalate to Human When
Qualify leads using business-defined criteria	Offer custom discounts or pricing outside approved ranges	Lead explicitly asks for a human sales rep
Answer standard pre-sales and pricing questions	Make final sales commitments or sign agreements	Negotiation or custom pricing is requested
Book meetings using approved calendar availability	Modify contract terms	High-value lead exceeds a configured deal-size threshold
Update CRM fields it owns (score, status, notes)	Access or promise data outside its permitted scope	Lead expresses frustration or confusion the employee cannot resolve
Notify the sales team of qualified leads		

Success Metrics

Success Metric	Why It Matters
Qualified Leads	Shows how many leads meet the business's fit criteria and are worth sales time.
Meetings Booked	Direct measure of pipeline generated by the AI employee.
Response Time	Speed of first response is one of the strongest predictors of lead conversion.
Conversion Rate	Tracks how many AI-qualified leads eventually become customers.

3.3 AI Customer Support Employee

Purpose

Repetitive support questions and slow response times damage customer experience and consume team hours that could go toward higher-value work. The AI Customer Support Employee exists to resolve common customer issues instantly and consistently, while knowing exactly when to bring in a human.

Responsibilities

- Answer frequently asked questions using business knowledge
- Resolve common issues (account, product, service questions)

- Track and share order status and delivery updates
- Handle complaints with an empathetic, on-brand tone
- Assist with return and refund requests within policy
- Escalate to a human when a request falls outside its authority

Workflow

Step	Stage	What Happens
1	**Customer Question**	A customer reaches out with a question, issue, or request through a connected channel.
2	**Knowledge Search**	The employee searches the business's approved knowledge base and policies for a relevant answer.
3	**Response**	A clear, on-brand answer is generated and sent to the customer.
4	**Action**	If the request requires an action (e.g., order lookup, return initiation), the employee performs it within its permitted scope.
5	**Resolution**	The issue is marked resolved, or escalated to a human with full context if it cannot be completed.

Business Rules

Can Do	Cannot Do	Escalate to Human When
Answer questions covered by the business knowledge base	Approve exceptions to refund or return policy	Customer explicitly requests a human agent
Look up and share order/delivery status	Make promises not covered by business policy	Issue is not covered by the knowledge base or policy
Initiate returns or refunds within defined policy limits	Access sensitive payment details	Complaint involves a policy exception or goodwill gesture
Log complaints and tag them by category	Close a complaint the customer disputes as unresolved	Customer sentiment indicates high frustration or escalation risk

Success Metrics

Success Metric	Why It Matters
Resolution Rate	Percentage of conversations fully resolved without human involvement.
Customer Satisfaction (CSAT)	Measures whether the support experience met customer expectations.
Response Speed	First-response and resolution time compared to human-only support.
Escalation Rate	Tracks how often the employee correctly identifies when to involve a human.

3.4 AI Receptionist

Purpose

Manual appointment booking is slow, error-prone, and often unavailable outside business hours, leading to missed opportunities and no-shows. The AI Receptionist exists to manage scheduling end-to-end, giving customers an always-available way to book, reschedule, or confirm appointments.

Responsibilities

- Book appointments based on real-time availability
- Manage the connected calendar on the business's behalf
- Check availability before confirming any time slot
- Reschedule or cancel meetings on request
- Send confirmations and reminders to reduce no-shows

Workflow

Step	Stage	What Happens
1	**Customer Request**	A customer asks to book, reschedule, or cancel an appointment.
2	**Availability Check**	The employee checks the connected calendar against working hours and existing bookings.
3	**Booking**	An available slot is offered and confirmed with the customer.
4	**Confirmation**	A confirmation is sent to the customer and the calendar is updated.
5	**Reminder**	Automated reminders are sent ahead of the appointment to reduce no-shows.

Business Rules

Can Do	Cannot Do	Escalate to Human When
Book within configured working hours and business rules	Book outside approved working hours or blocked-off time	No suitable time slot is available and the customer needs flexibility
Reschedule or cancel existing appointments on request	Override double-booking or capacity limits	Customer requests an exception to booking rules
Send confirmations and reminders automatically	Modify appointment types or pricing tied to a booking	Repeated rescheduling suggests a service issue
Check and communicate real-time availability	Guarantee a specific staff member unless configured	Customer explicitly asks to speak with staff directly

Success Metrics

Success Metric	Why It Matters
Appointments Booked	Volume of scheduling handled without manual coordination.
No-Show Reduction	Impact of automated reminders on attendance rates.
Response Time	Speed from request to confirmed booking.
Rebooking Rate	Indicates how well cancellations and reschedules are recovered.

3.5 AI Follow-up Employee

Purpose

Deals stall and customers disengage when follow-up is inconsistent or forgotten. The AI Follow-up Employee exists to keep leads, prospects, and customers engaged at the right moments, without manual tracking by the team.

Responsibilities

- Follow up with leads who have gone quiet
- Send proposal and quote reminders
- Re-engage inactive or at-risk customers
- Recover abandoned carts for e-commerce businesses
- Run structured follow-up campaigns over email and WhatsApp

Workflow

Step	Stage	What Happens
1	**Event Trigger**	A trigger occurs — no response after X days, an unpaid proposal, an abandoned cart, or a scheduled re-engagement.
2	**Analyze Situation**	The employee reviews the customer's history and prior interactions to decide the right approach.
3	**Generate Message**	A relevant, on-brand follow-up message is generated for the situation.
4	**Send**	The message is sent through the appropriate channel (email or WhatsApp).
5	**Track Result**	The response (or lack of one) is tracked and used to plan the next follow-up or end the sequence.

Business Rules

Can Do	Cannot Do	Escalate to Human When
Send follow-ups based on configured triggers and timing	Exceed configured frequency or contact limits	Customer responds with a complex question outside follow-up scope
Personalize messages using known customer context	Contact customers who have opted out	Customer expresses frustration about being contacted
Run multi-step follow-up sequences within set limits	Offer discounts or incentives not pre-approved	A high-value deal shows renewed interest and needs sales attention
Stop a sequence when the customer responds or converts	Continue a sequence after an explicit stop request	

Success Metrics

Success Metric	Why It Matters
Conversion Improvement	Additional bookings, sales, or recoveries driven by follow-up.
Response Rate	Share of follow-up messages that get a reply.
Retention	Impact of re-engagement on repeat customers over time.

3.6 Shared AI Employee Capabilities

Every AI Employee is built on a common set of product-level capabilities. These are described here at a product level; how they are implemented technically is covered in the architecture section of this PRD.

Natural Conversations

Each employee communicates in clear, on-brand, human-like language appropriate to its role, avoiding robotic or scripted phrasing.

Business Knowledge Understanding

Employees are grounded in the specific business's information — products, services, policies, and FAQs — rather than generic knowledge.

Memory

Employees remember prior interactions with a customer, so conversations continue naturally instead of starting over each time.

Context Awareness

Employees understand where a customer is in their journey (new lead, existing customer, past complaint) and respond accordingly.

Decision Making

Within their defined business rules, employees decide the appropriate next action — answer, act, or escalate — without needing manual input for every step.

Tool Usage

Employees use approved business tools and integrations (calendar, CRM, messaging channels) to complete real tasks, not just produce text.

Human Handoff

Every employee recognizes the limits of its authority and hands off to a human teammate smoothly, with full context, whenever a situation requires it.

3.7 AI Employee Interaction Model

Business Owner ↔ AI Employees

Business owners and their teams configure each AI Employee's rules, working hours, tone, and permitted actions, and monitor their activity and outcomes from the business workspace. Employees keep the business informed through notifications, summaries, and CRM updates rather than requiring constant supervision.

Customer ↔ AI Employees

Customers interact with AI Employees the same way they would with a real team member — through chat, WhatsApp, email, or web forms — without needing to know an AI is involved unless the business chooses to disclose it. The customer experience should feel like dealing with a responsive, competent employee, not filling out a form.

Information Sharing Across Employees and the Business

Because all AI Employees share the same workspace, updates made by one employee are immediately visible to the others and to the business. A lead qualified by the AI Sales Employee is instantly available to the AI Receptionist for booking and to the AI Follow-up Employee if no meeting is booked.

Example Scenarios

- A website visitor chats with the AI Sales Employee, gets qualified as a Hot lead, and books a meeting — the AI Receptionist confirms the slot and sends reminders automatically.
- A customer messages on WhatsApp asking where their order is — the AI Customer Support Employee looks up the order and responds within seconds.
- A prospect goes quiet after receiving a proposal — the AI Follow-up Employee sends a reminder three days later and notifies sales if the prospect re-engages.
- A customer asks for a refund outside policy — the AI Customer Support Employee explains what it can do and escalates the exception to a human teammate with full context.

3.8 Future AI Employees

The following roles are directional and out of scope for the MVP. They are included to show how the AI Workforce is expected to expand.

Future Employee	High-Level Description
AI HR Employee	Supports hiring workflows, candidate screening, onboarding tasks, and internal employee questions.
AI Marketing Employee	Assists with campaign content, social posting cadence, and basic performance reporting.
AI Finance Employee	Helps with invoicing, payment reminders, and basic financial reporting for the business.
AI Operations Employee	Coordinates internal workflows, task tracking, and cross-team operational reminders.

These roles will follow the same product model established in this section: a clear purpose, defined responsibilities, a documented workflow, explicit business rules, and measurable success metrics.

Section 4: Technical Architecture & System Requirements

Prepared as a standalone section for integration into the master PRD. Audience: engineering, architecture, and technical stakeholders.

4.1 Technical Overview

Trevolk AI Workforce is a multi-tenant SaaS platform, not a single chatbot product. Each business workspace can activate one or more specialized AI Employees (Sales, Customer Support, Receptionist, Follow-up) that operate as independent functional agents while running on a single shared technical foundation: one frontend, one backend, one authentication system, one database, and one AI engine. This "one platform, many agents" model is the central architectural constraint for every decision in this section.

The system is organized into five cooperating layers:

- Frontend Layer — the dashboard and workspace UI used by business owners and their teams to configure AI Employees, monitor conversations, and manage leads, customers, and appointments.
- Backend API Layer — the central service that owns business logic, authentication, authorization, and orchestration between the frontend, the AI engine, the database, and external integrations.
- AI Agent Engine — the reasoning layer that powers each AI Employee: prompt construction, LLM calls, tool invocation, and conversation memory.
- Database Layer — the system of record for all workspace, employee, conversation, customer, and operational data.
- Integration Layer — connectors to external channels and tools (WhatsApp, Gmail, Google Calendar, Stripe, Slack, Shopify, HubSpot) that let AI Employees act in the real world.

This layered separation fits an AI SaaS platform for three practical reasons. First, it isolates the parts of the system that change fastest — prompts, agent logic, and LLM providers — from the parts that must stay stable, such as authentication and the core data model, so AI experimentation never puts tenant data at risk. Second, it keeps every AI Employee thin: an employee is a configuration (role, tools, workflow, permissions) rather than a separate codebase, which is what makes "add a new AI Employee type" a data and prompt change instead of a new application. Third, a shared backend and database with strict workspace scoping is the simplest way to guarantee multi-tenant data isolation while still allowing the platform to add new integrations and employees without re-architecting the system.

4.2 Recommended Technology Stack

The stack below follows the technology choices already defined in the Master Context Document. Each choice is explained in terms of why it fits an early-stage, AI-heavy, multi-tenant SaaS product rather than a large-scale enterprise build.

Frontend

Aspect	Choice	Why
Framework	Next.js (React)	Server-side rendering and file-based routing give fast initial loads for a dashboard-heavy product and simplify SEO for public/marketing pages.
Language	TypeScript	Type safety across a growing set of AI Employee configurations, API contracts, and shared UI reduces runtime bugs as the codebase scales.

Aspect	Choice	Why
Styling	Tailwind CSS	Utility-first styling supports the minimal, enterprise-grade design language (dark charcoal background, blue accents, soft rounded corners) without a heavy custom CSS layer.
Component strategy	Reusable, composable component library	Shared primitives (cards, tables, modals, chat panels, skeleton loaders) are built once and reused across every AI Employee dashboard, keeping the UI consistent as new employees are added.

Table 4.1 — Frontend stack summary

Backend

Aspect	Choice	Why
Framework	Node.js with Express.js (NestJS optional as the codebase matures)	Node.js keeps the frontend and backend in one language (TypeScript), which speeds up an early-stage team. Express is lightweight for MVP; NestJS's module system can be adopted later if the codebase needs stronger structure.
API architecture	REST API, versioned (e.g. /api/v1/...)	REST is simple to document, test, and consume from the dashboard and future mobile/integration clients. Versioning protects existing integrations as the API evolves.
Service organization	Layered: routes → controllers → services → agents/integrations	Separating HTTP handling from business logic and from AI/agent logic keeps each concern independently testable and lets the AI layer evolve without touching controllers.

Table 4.2 — Backend stack summary

Database

Aspect	Choice	Why
Database	PostgreSQL	A relational database enforces the strong consistency and relationships needed between businesses, workspaces, employees, customers, leads, and appointments — all core entities in this platform.
ORM	Prisma	Type-safe queries and migrations that match the TypeScript backend, with a schema file that doubles as living documentation of the data model.
Storage approach	Single multi-tenant database, workspace-scoped by a workspace/business ID on every relevant table	Simpler to operate and cheaper than per-tenant databases at MVP scale, while still enforcing isolation at the query layer. A vector database can be added later for long-term semantic memory.

Table 4.3 — Database stack summary

Authentication

Aspect	Choice	Why
User authentication	Clerk or Supabase Auth	Managed authentication removes the cost of building and securing password handling, session management, and social login from scratch — important for a lean early-stage team.
Workspace access	Every user session is scoped to one or more workspaces (businesses)	Matches the multi-tenant model: a user can belong to multiple businesses, and every API call must resolve which workspace it applies to.
Roles and permissions	Role-based access control at the workspace level (e.g. Owner, Admin, Team Member)	Business owners need to delegate day-to-day monitoring to sales or support staff without giving them full account control.

Table 4.4 — Authentication approach

AI Layer

Aspect	Choice	Why
LLM providers	OpenAI, Gemini, Claude, Groq (provider-agnostic)	Supporting multiple providers behind one interface avoids vendor lock-in, allows cost/performance tuning per AI Employee, and provides a fallback if one provider has an outage.
Agent communication	Backend calls a common AI Agent Engine interface, which routes to the correct provider and agent configuration	Keeps the backend decoupled from any single LLM API, so switching or adding a provider does not require touching business logic.
Prompt management	Centralized prompt/template store per AI Employee type, versioned	Prompts change frequently as behavior is tuned; centralizing and versioning them avoids scattering prompt text through the codebase and allows safe rollback.

Table 4.5 — AI layer approach

Infrastructure

Aspect	Choice	Why
Deployment	Docker containers on a VPS or cloud provider	Containerization keeps environments consistent between development and production and makes it straightforward to move providers or scale horizontally later.
Storage	Managed object storage for files/attachments (e.g. knowledge base documents, media)	Keeps large binary content out of PostgreSQL and allows independent scaling of file storage.
Monitoring	Centralized application logging, error tracking, and uptime monitoring	Early visibility into failures (especially AI/LLM call failures and integration errors) is essential when the product depends on third-party APIs.

Table 4.6 — Infrastructure approach

4.3 High-Level System Architecture

At a high level, a request flows through the platform in one direction, with responses and events flowing back:

Frontend → Backend API Layer → AI Agent Engine → Database → External Integrations

Figure 4.1 — Primary request flow. Integrations and the database also communicate back up through the Backend API Layer to the Frontend (e.g. new conversation messages, appointment confirmations).

Frontend

Renders the workspace dashboard, AI Employee configuration screens, conversation views, and reporting. It never talks to the database or the AI providers directly — every action goes through the Backend API Layer, which keeps a single, auditable entry point into the system.

Backend API Layer

Owns authentication checks, workspace/tenant resolution, request validation, and business rules (e.g. lead creation, appointment scheduling rules). It is the only layer allowed to talk to the database directly and the only layer that invokes the AI Agent Engine or external integrations.

AI Agent Engine

Receives a task from the backend (e.g. "respond to this customer message as the Support Employee for Workspace X"), assembles the right prompt and context, calls the selected LLM provider, optionally invokes tools (CRM update, calendar lookup, order lookup), and returns a structured result to the backend.

Database

Persists everything the platform needs to remember: workspaces, AI Employee configurations, conversations, customers, leads, appointments, and integration credentials. It is written to and read from exclusively through the backend.

External Integrations

Channels and third-party tools the AI Employees act through — WhatsApp and Gmail for communication, Google Calendar for scheduling, Stripe for billing, Slack for internal notifications, and Shopify/HubSpot for commerce and CRM data. Integrations are invoked by the backend (directly, or via the AI Agent Engine's tool layer) and never called from the frontend.

4.4 Application Structure

The codebase is organized as a single frontend application and a single backend application, each internally modular. This avoids duplicate code across AI Employees while keeping each employee's logic easy to locate and extend.

Frontend Structure

- **pages/:** Route-level entry points (Next.js routing) — dashboard, workspace settings, AI Employee views, login/onboarding.
- **components/:** Shared, reusable UI primitives (buttons, cards, tables, modals, chat bubbles, skeleton loaders) used across every screen.
- **features/:** Feature-specific modules grouped by domain (e.g. features/sales-employee, features/support-employee, features/appointments) each containing its own components, state, and API calls.
- **hooks/:** Reusable React hooks for data fetching, authentication state, workspace context, and real-time updates.
- **utils/:** Shared helper functions — formatting, validation, constants, API client wrappers.

Backend Structure

- **routes/:** Defines API endpoints and maps them to controllers; organized by domain (auth, employees, conversations, leads, appointments, integrations).
- **controllers/:** Handle HTTP request/response concerns — input validation, calling the right service, formatting the response.
- **services/:** Contain business logic — lead qualification rules, appointment rules, workspace management — independent of HTTP or database detail.
- **agents/:** AI Employee logic: one module per employee type (sales, support, receptionist, follow-up), each defining its prompts, tools, and behavior on top of the shared AI Agent Engine.
- **integrations/:** One connector module per external service (WhatsApp, Gmail, Google Calendar, Stripe, Slack, Shopify, HubSpot), each exposing a consistent interface to the services and agents layers.
- **database/:** Prisma schema, migrations, and data-access utilities.
- **auth/:** Authentication middleware, session/workspace resolution, and role/permission checks.

This structure keeps each AI Employee's logic isolated inside its own module in agents/ while all employees continue to share the same routes, services, database layer, and integrations — directly matching the "one platform, many agents" principle from the Master Context Document.

4.5 AI Engine Architecture Overview

The AI Agent Engine is the shared reasoning core that every AI Employee runs on. Rather than building a separate AI system per employee, each AI Employee is a configuration on top of one engine:

- **Shared AI engine:** A single service handles prompt assembly, LLM provider calls, response parsing, and tool execution for all AI Employees. This avoids duplicating LLM integration code four (and eventually more) times.
- **Agent separation:** Each AI Employee type (Sales, Support, Receptionist, Follow-up) is defined by its own role definition, prompt template, permitted tools, and business rules, kept in the agents/ module. Adding a new employee type means adding a new configuration, not a new engine.
- **Tool usage:** AI Employees act on the world through a defined set of tools (e.g. "update CRM", "check calendar availability", "look up order status") rather than free-form access to the database or integrations. This keeps AI actions predictable, auditable, and limited to what each employee is actually responsible for.
- **Memory:** Each AI Employee retains conversation-level memory (recent message history) and workspace-level context (business rules, FAQs, past interactions with a given customer) so responses stay consistent across a conversation and across sessions. Memory is currently stored in PostgreSQL, with a vector database planned for future long-term semantic recall.
- **Context handling:** Before each AI response, the engine assembles the relevant context — the AI Employee's role and rules, recent conversation history, relevant customer/lead data, and knowledge base content — into the prompt, so the model always answers with workspace-specific and conversation-specific context rather than generic knowledge.

4.6 Database Overview

The data model is designed around a clear tenancy hierarchy — a Business owns one or more Workspaces, and everything else (AI Employees, conversations, customers, leads, appointments) is scoped underneath. Full schemas are out of scope for this section; the entities and relationships below define the shape of the model.

Entity	Purpose
Users	People who can log in — owners, admins, and team members. A user can belong to one or more workspaces.
Businesses / Workspaces	The tenant boundary. Every other entity is scoped to a workspace, which enforces data isolation between customers of the platform.
AI Employees	Instances of an AI Employee type activated within a workspace, holding its configuration, permissions, and status.
Conversations	Message threads between an AI Employee and a customer/lead, across channels (chat, WhatsApp, email).
Customers	End customers of a business, referenced by conversations, orders, and support tickets.
Leads	Prospective customers captured by the AI Sales Employee, with qualification status and CRM linkage.
Appointments	Scheduled meetings managed by the AI Receptionist, linked to a customer/lead and a calendar integration.
Knowledge Base	Business-specific reference content (FAQs, policies, product info) used by AI Employees to answer questions accurately.
Integrations	Stored credentials/configuration for connected external services (WhatsApp, Gmail, Calendar, Stripe, Slack, Shopify, HubSpot) per workspace.
Analytics	Aggregated or event-level data used for reporting on AI Employee performance, response times, and conversions.

Table 4.7 — Core data entities

At a high level: a Business has many Workspaces (or is treated as a single workspace for MVP simplicity); a Workspace has many AI Employees, Customers, Leads, Appointments, Knowledge Base entries, and Integrations; Conversations belong to one AI Employee and one Customer/Lead; Appointments belong to one Customer/Lead and are created through the Receptionist employee; Analytics records reference the workspace and, where relevant, a specific AI Employee or conversation. Every table that stores tenant data carries a workspace identifier so queries and access checks can enforce isolation consistently.

4.7 API Requirements Overview

The backend exposes a versioned REST API grouped by domain. Each group is described by its purpose, not its exact endpoints, since detailed contracts belong in a separate API specification.

API Category	Purpose
Authentication APIs	Sign-up, login, session/token handling, and workspace resolution for the authenticated user.
AI Employee APIs	Activate, configure, update, and deactivate AI Employees within a workspace.
Conversation APIs	Retrieve, send, and update messages within a conversation; power real-time chat views in the dashboard.
Customer APIs	Create, look up, and update customer records referenced across support, sales, and appointments.
Lead APIs	Create and manage leads, qualification status, and CRM sync triggered by the AI Sales Employee.

API Category	Purpose
Appointment APIs	Create, reschedule, cancel, and list appointments; sync with connected calendars.
Integration APIs	Connect, configure, and disconnect third-party services (WhatsApp, Gmail, Calendar, Stripe, Slack, Shopify, HubSpot) per workspace.
Analytics APIs	Retrieve aggregated metrics on AI Employee activity, response times, conversions, and workspace usage.

Table 4.8 — API categories and their purpose

4.8 Security Requirements

- **Authentication security:** All authentication is handled through the managed provider (Clerk/Supabase Auth), with secure session/token handling, enforced password policies, and support for social login where offered by the provider.
- **Authorization:** Every API request resolves the authenticated user's workspace and role before executing, so users can never act outside a workspace they belong to.
- **Data privacy:** Customer, lead, and conversation data belonging to one workspace is never accessible to another workspace; workspace scoping is enforced at the database query layer, not only in the UI.
- **API security:** All API traffic runs over HTTPS, with rate limiting on public and authentication endpoints, and input validation on every request to prevent malformed or malicious payloads.
- **User permissions:** Role-based access (Owner, Admin, Team Member) restricts sensitive actions — such as connecting integrations, billing, or deleting an AI Employee — to authorized roles only.
- **Secure integrations:** Credentials and tokens for third-party integrations (WhatsApp, Gmail, Stripe, etc.) are encrypted at rest and never exposed to the frontend; all calls to these services are made server-side.

4.9 Scalability Considerations

The MVP architecture is intentionally simple, but the following considerations keep it from requiring a rewrite as usage grows:

- **Multiple businesses:** Workspace-scoped data access already supports many tenants on shared infrastructure; heavier tenants can later be moved to dedicated resources if needed without changing the data model.
- **Multiple AI Employees:** Because employees are configurations on a shared engine, adding new employee types (HR, Marketing, Finance, Operations, Recruitment) is additive and does not require new infrastructure.
- **Increasing users:** Stateless backend services behind a load balancer can be scaled horizontally as concurrent users grow; the database can move to read replicas if read load becomes a bottleneck.
- **Background tasks:** Long-running or delayed work — follow-up sequences, reminders, abandoned cart recovery, report generation — should run through a background job/queue system rather than inside the request/response cycle, keeping the API responsive.
- **More integrations:** The integrations/ module pattern means each new connector is added independently, with a consistent interface, without touching unrelated parts of the backend.

None of these require premature investment for MVP — they describe the extension points the current architecture already supports.

4.10 Development Principles

- **Clean code:** Consistent naming, small focused functions, and clear separation between HTTP, business logic, and AI/agent logic across the codebase.
- **Reusable components:** Shared UI components and backend services are built once and reused across AI Employees rather than duplicated per employee.
- **Modular architecture:** Each AI Employee, integration, and major feature lives in its own module with a clear boundary, so features can be added or modified without ripple effects elsewhere.
- **Documentation:** API contracts, database schema, and each AI Employee's prompt/behavior configuration are documented and kept up to date as the source of truth for engineering decisions.
- **Testing:** Core business logic (lead qualification, appointment rules, permission checks) and integration connectors are covered by automated tests; AI agent responses are validated through structured checks in addition to manual review.
- **Maintainability:** New engineers should be able to trace a request from frontend to backend to AI engine to database without hidden coupling, so the platform can keep adding AI Employees and integrations without accumulating technical debt.

End of Section 4

Section 5: MVP Roadmap, Launch Strategy & Future Scaling

5.1 MVP Definition

What is the first version of Trevolk AI Workforce?

The MVP is a single-tenant-ready, multi-tenant-capable SaaS dashboard where a business signs up, activates one AI Employee, connects a small number of core channels and data sources, and puts that AI Employee to work handling real conversations within the same day. The MVP is not the full “AI Workforce” vision with five or more coordinated employees — it is the smallest slice of that vision that is genuinely usable and genuinely sellable: one workspace, one AI Employee, one clear job done well.

What problem should the MVP solve?

Per the Master Context Document, the sharpest and most universal pain points across the primary target markets (e-commerce, digital agencies, software houses, coaches/educators, real estate) are slow response times, repetitive customer questions, and lost/unfollowed leads. The MVP should prove that Trevolk can take over a single, well-defined, repetitive workflow — end to end — more reliably and more cheaply than a human doing it manually, without requiring the business to change how it already operates.

What features are absolutely necessary?

- Business signup, workspace creation, and authentication (single workspace per business at launch).
- Activation of one AI Employee with a guided setup flow (business info, tone, FAQs/knowledge, escalation rules).
- A working chat channel (embeddable website widget and/or WhatsApp) that the AI Employee can actually respond on.
- A simple dashboard: conversation history, basic analytics, and a way to review/correct AI answers.
- Human handoff / escalation path when the AI cannot confidently resolve a request.
- Basic subscription billing (Stripe) so the product can be sold from day one, not just demoed.

What should be excluded from MVP?

- Multiple AI Employees running simultaneously for one business.
- Deep two-way integrations (full CRM sync, Shopify catalog sync, Google Calendar two-way sync, HubSpot, Slack).
- Vector-database long-term memory; MVP uses simpler structured + short-term memory.
- Enterprise features: SSO, custom roles/permissions, SLAs, white-labeling.
- A public AI Employee marketplace.
- Voice/phone channel support.

Guiding principle: build one thing a small team can ship in weeks, that a real business will pay for, and that generates enough usage data and feedback to justify building the second AI Employee.

5.2 MVP Feature Prioritization

Priorities are set using a simple test: does this feature let a real business get real value in week one? If yes, it is Must Have. If it meaningfully improves retention or conversion but the product still works without it, it is Should Have. If it only matters once there are multiple paying customers or multiple AI Employees, it is Later.

Priority	Feature	Why
Must Have	Signup, login, workspace, billing (Stripe)	No sellable product without accounts and payment.
Must Have	One AI Employee (Support or Sales) with setup wizard	This is the core value proposition; everything else supports it.
Must Have	Website chat widget	Fastest, lowest-friction channel to deploy and demo.
Must Have	Conversation dashboard + transcript log	Businesses need to see and trust what the AI is doing.
Must Have	Human escalation / handoff	Prevents bad outcomes and builds trust during early trust-building phase.
Must Have	Basic analytics (volume, resolution rate)	Minimum proof of value needed to justify the subscription.
Should Have	WhatsApp channel	High-value for target markets but adds integration complexity.
Should Have	Second AI Employee (e.g., Sales or Receptionist)	Expands value once the first employee is validated.
Should Have	Email notifications / daily summary	Improves retention without being core to the workflow.
Should Have	Basic CRM/Calendar integration (one-way)	Useful but not required to prove the core loop works.
Later	Multi-employee orchestration in one workspace	Only matters once businesses already trust a single employee.
Later	Vector-based long-term memory	Optimization, not a launch blocker.
Later	Enterprise features (SSO, roles, SLAs)	Irrelevant until enterprise deals are actually in the pipeline.
Later	AI Employee marketplace	A future-vision feature, not an MVP concern.

5.3 Development Roadmap

The roadmap below assumes a small founding team (roughly 2–4 people: one product-focused founder, one to two engineers, and a part-time designer/growth person). Timelines are indicative, not committed, and should be re-validated after each phase based on real customer feedback.

Phase 1: Foundation

Goal: Stand up the core platform shell so an AI Employee has somewhere to live.

Features: Authentication and workspace creation; base dashboard shell and navigation; billing integration (Stripe) in test mode; core data model for businesses, users, and conversations; design system applied per Section 3 (dark charcoal + blue palette, minimal glass effects).

Expected outcome: A business can create an account, log in, and land on an empty but functional dashboard. Nothing customer-facing yet — this phase is purely infrastructure.

Phase 2: First AI Employee

Goal: Build, launch, and validate a single AI Employee with a small group of real early customers.

Features: Setup wizard for the chosen employee (business info, FAQs/knowledge base, tone/rules, escalation triggers); website chat widget; conversation transcript log and basic analytics; human handoff flow; first paid plan available for purchase.

Expected outcome: 5–10 paying or actively piloting businesses using the AI Employee on real conversations, with enough usage to identify what breaks, what confuses users, and what feature is missing most.

Phase 3: Additional AI Employees

Goal: Prove the modular "AI Workforce" model by adding a second AI Employee without rebuilding the platform.

Features: Second AI Employee module (built on shared infrastructure per the architecture principles in Section 2/4); WhatsApp channel; one-way CRM or calendar integration; upgraded analytics comparing employees; referral or upsell flow for existing customers.

Expected outcome: Existing customers activate a second employee, validating cross-sell; new customers can choose which employee to start with based on their biggest pain point.

Phase 4: Business Scaling

Goal: Move from "works for early adopters" to "reliable enough to scale sales and marketing."

Features: Deeper integrations (Shopify, HubSpot, Google Calendar two-way, Slack); reliability and monitoring improvements; self-serve onboarding (reducing manual founder-led setup); expanded plans including an Enterprise tier; groundwork for future AI Employees and eventually a marketplace.

Expected outcome: The product can be sold and onboarded with minimal founder involvement, supporting a repeatable growth motion instead of hands-on pilots.

5.4 Recommended MVP Strategy

Which AI Employee should be built first, and why

Build the AI Customer Support Employee first. Reasoning:

- Lowest integration complexity: it primarily needs a knowledge base and conversation logic, not deep CRM or calendar sync (unlike the AI Sales Employee or AI Receptionist).

- Universal pain point: “repetitive customer questions” and “slow response times” apply to every primary target segment, not just one vertical.
- Fast, visible ROI: a business can see, within days, how many questions were answered automatically and how much time that saved — an easy story to sell and renew on.
- Natural upgrade path: once support is trusted, the same relationship makes it easy to sell the AI Sales Employee or AI Follow-up Employee next.

Which target market should be approached first, and why

Start with e-commerce businesses. They have high, predictable volumes of repetitive questions (order status, returns, product questions), a clear willingness to pay for tools that protect revenue, and straightforward channels to reach them (Shopify community, e-commerce Facebook/WhatsApp groups, agencies that serve them). Digital agencies and software houses are a strong secondary channel because they can resell or recommend Trevolk to their own client base, multiplying reach without direct customer acquisition cost.

Why this approach reduces risk

- One employee, one workflow → smaller surface area to build, test, and debug.
- One clear market → focused messaging, easier to find and talk to prospects, faster feedback loops.
- Avoids the classic early-stage trap of building five AI Employees for five markets simultaneously and validating none of them well.

How to validate customer demand

- Run a small number of manual, high-touch pilots (5–10 businesses) before any self-serve signup exists.
- Track a simple validation signal: would this business pay to keep using it after the pilot, and did it measurably reduce response time or ticket volume?
- Use pilot conversations directly to refine the setup wizard and FAQ/knowledge structure before opening broader signups.

5.5 Customer Acquisition Strategy (Initial)

Finding first customers

- Direct outreach to e-commerce store owners and agencies in existing founder networks and relevant online communities (Shopify groups, e-commerce Facebook/LinkedIn groups, local business networks).
- Partner with 2–3 digital agencies or software houses willing to introduce Trevolk to their client base in exchange for early access or a referral incentive.
- Build in public: share build progress, demos, and results on LinkedIn/X to attract inbound interest from the target audience.

Demo strategy

- Offer a short, live, personalized demo using the prospect's own FAQs or sample product catalog rather than a generic script — make the value obvious in minutes.
- Follow every demo with a time-boxed free pilot (e.g., 14–21 days) on their real website traffic, not a sandbox.

Early feedback collection

- Weekly short check-ins with each pilot customer during the trial period.
- In-dashboard feedback prompt after key actions (e.g., reviewing a conversation) to capture friction in the moment.

- Track every conversation the AI could not resolve as a direct input into the product backlog.

Customer onboarding

- Founder-led, hands-on onboarding for the first cohort of customers — this is expected and valuable at MVP stage, not a shortcut to avoid.
- A simple, guided setup wizard reduces this manual burden progressively as the product matures (see Phase 4).

Product improvement cycle

- Short, weekly iteration cycles during the pilot phase: ship fixes and small improvements based on that week's real conversations.
- Formal monthly review of usage data and feedback to decide what moves from "Should Have"/"Later" into the active roadmap.

5.6 Pricing Strategy Overview

Exact price points are intentionally not fixed at this stage — they should be set and tested against real willingness-to-pay during pilots. The structure below defines the shape of the pricing, not final numbers.

Plan	Intended For	Illustrative Scope
Starter	Small businesses / solo operators trying their first AI Employee	1 AI Employee, 1 channel (website widget), capped monthly conversation volume, basic analytics
Business	Growing e-commerce, agencies, and coaches with active customer volume	1–2 AI Employees, multiple channels (website + WhatsApp), higher/uncapped conversation volume, integrations, priority support
Enterprise	Larger businesses or agencies managing multiple clients/workspaces	Multiple AI Employees, custom integrations, dedicated onboarding, advanced reporting, SLA-backed support (built out in Phase 4)

What determines pricing

- Number of active AI Employees enabled for the workspace.
- Conversation/interaction volume per month.
- Number of connected channels and integrations.
- Level of support and onboarding provided.

How value is communicated

Pricing should be framed around cost and time saved rather than technology features — e.g., positioning the plan against the cost of a part-time support or sales hire, and against the revenue lost to slow responses and missed follow-ups. Dashboards should make the "time saved" and "conversations handled" numbers visible so the value is self-evident at renewal time.

Future monetization opportunities

- Usage-based add-ons for high-volume businesses beyond plan limits.
- Premium integrations (e.g., advanced CRM/Shopify sync) as paid add-ons.
- A future AI Employee marketplace with revenue share on third-party or specialized employees.

- Agency/reseller pricing tier for partners managing multiple client workspaces.

5.7 Success Metrics

Product Metrics

Metric	What it Tells Us
Active businesses (weekly/monthly)	Whether businesses keep coming back to use the product, not just sign up.
AI Employee activation rate	How many signups actually complete setup and go live.
Conversations handled per business	Real usage depth, not just account creation.
AI resolution rate (no human handoff needed)	Whether the AI Employee is actually doing the job, not just chatting.

Business Metrics

Metric	What it Tells Us
New paying customers per month	Whether acquisition efforts are working.
Monthly recurring revenue (MRR)	Overall business health and growth rate.
Customer retention / churn rate	Whether the product delivers ongoing value, not just a good first impression.
Pilot-to-paid conversion rate	Whether free trials/pilots are translating into real revenue.

Customer Value Metrics

Metric	What it Tells Us
Average response time (before vs. after)	Direct evidence of the core problem being solved.
Estimated time saved per week per business	Tangible ROI businesses can point to internally.
Leads captured / follow-ups completed	Value delivered specifically by sales- and follow-up-oriented employees.
Customer satisfaction on AI-handled conversations	Whether automation is improving, not hurting, customer experience.

5.8 Future Expansion

- More AI Employees: progressively introduce the AI Sales Employee, AI Receptionist, and AI Follow-up Employee, followed by longer-term roles such as AI HR, AI Marketing, AI Finance, AI Operations, and AI Recruitment Employees.
- More integrations: expand beyond the MVP channel set to Gmail, Google Calendar, Slack, Shopify, and HubSpot as outlined in the technology stack.
- AI Employee marketplace: allow businesses to discover, add, and eventually customize specialized AI Employees beyond the core lineup.

- Enterprise features: SSO, custom roles and permissions, SLAs, and white-labeling for larger organizations and agency partners.
- Advanced automation: cross-employee workflows where, for example, the AI Sales Employee hands a qualified lead directly to the AI Receptionist to book a meeting, and to the AI Follow-up Employee for nurture — the true "AI Workforce" collaborating as a team.
- Expansion into future verticals named in the Master Context: clinics, restaurants, SMEs, and larger enterprises.

5.9 Final Product Direction

Trevolk AI Workforce aims to become the default way small and mid-sized businesses staff their repetitive operations — not through a single chatbot, but through a coordinated team of specialized AI Employees operating on one unified, enterprise-grade platform. Its long-term potential comes from the modular architecture defined in this document: because every AI Employee shares one frontend, backend, authentication system, database, and AI engine, each new employee adds value to the whole platform rather than starting from zero, and each new customer segment strengthens the same core product instead of fragmenting it.

Core principles for future development:

- Validate before expanding — no new AI Employee or market is pursued without evidence from real usage and real customers.
- Ship the smallest valuable slice — prioritize speed to a usable, sellable product over completeness.
- Preserve the unified architecture — growth in employees, integrations, and markets should never require duplicating core infrastructure.
- Let customer feedback drive the roadmap — pricing, features, and prioritization all trace back to real business outcomes: time saved, leads captured, and experience improved.